UDACITY
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SCHOOL OF PROGRAMMING AND DEVELOPMENT

SQL

Syllabus

Overview

Master SQL, the essential language for data analysis and insight-driven decision-making. This program covers SQL fundamentals, advanced querying, and database management. Learn to work with multiple tables, perform data aggregation and cleaning, use window functions, and optimize queries for large datasets. Apply your skills to real-world projects, analyzing deforestation data and redesigning databases. Gain proficiency in SQL for data analysis, database design, and management, preparing for roles requiring strong data manipulation and insight generation. Ideal for aspiring data analysts, business intelligence professionals, and those seeking to enhance their data querying and management skills.

Nanodegree Program | Beginner | 41 hours | 4.6 (292 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

Basic SQL

You will also need to be able to communicate fluently and professionally **in written and spoken English**.

Skills You'll Learn

Introduction to SQL

SQL subqueries | SQL joins | SQL aggregations | SQL window functions | SQL queries | Data cleaning | SQL query performance tuning

Management of Relational and Non-relational Databases

Mongodb | SQL CRUD commands | Database normalization | Redis | Database indexing | Data manipulation language | Query plans | Database schemas | SQL query performance tuning | Rdbms performance and optimization | Foreign keys | Database fundamentals | Business rule enforcement in databases

Courses

01. Welcome to the SQL Nanodegree Program

40 minutes

Welcome to the SQL Nanodegree program! Learn more about the pre-requisites, structure of the program, and getting started!

L1	SQL Nanodegree Program Introduction	Welcome to the SQL Nanodegree program! In this lesson, you will learn more about the structure of the program and meet your instructors.
L2	Getting Help	You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content.

02. Introduction to SQL

18 hours

SQL is one of the most versatile tools available for extracting insights from stored data.
Learn how to execute core SQL commands to define, select, manipulate, control access, aggregate, and join data and data tables. Understand when and how to use subqueries, several window functions, and partitions to complete complex tasks. Clean data, optimize SQL queries and write select advanced JOINs to enhance analysis performance, explain which cases you would want to use particular SQL commands, and apply the results from queries to address business problems.

L1	SQL Operators Introduction	An Introduction to the course and using the Parch & Posey sales database to solve real-world business questions.
L2	Introduction to SQL	Explore SQL basics, its importance for data analysis, business use cases, database structure, and an overview of popular SQL databases with hands-on practice.
L3	Entity Relationship Diagrams	Learn how to interpret Entity Relationship Diagrams (ERDs) to visualize tables, columns, and relationships in a relational database.
L4	An Introduction to Statements in SQL	Learn SQL basics by understanding key statements like SELECT and FROM, practicing queries, and following best practices for formatting and execution in a SQL workspace environment.
L5	The LIMIT Clause in SQL	Learn how to use the LIMIT clause in SQL to restrict the number of rows returned in query results, making data exploration faster and more manageable.

L6	ORDER BY Clause in SQL	Learn to use the SQL ORDER BY clause to sort query results by one or multiple columns, in ascending or descending order, with practical query exercises and solutions.
L7	WHERE Clause in SQL	Learn to use the SQL WHERE clause to filter table data based on numeric or text conditions, using operators like =, !=, >, <, and practice with real-world queries.
L8	Arithmetic Operators in SQL	Learn how to use arithmetic operators in SQL to create derived columns, perform calculations in queries, and apply order of operations for accurate results.
L9	LIKE and IN Operators in SQL	Learn how to use SQL's LIKE and IN operators to filter text and numeric data efficiently, enabling flexible database queries with wildcards and value lists.
L10	AND and BETWEEN Operators in SQL	Learn to use AND and BETWEEN operators in SQL to filter data with multiple conditions and ranges, practicing queries on real tables and understanding inclusivity of endpoints.
L11	NOT and OR Operators in SQL	Learn how to use SQL's NOT and OR operators to filter data, combine conditions, and query for records that do not or do match given criteria, with practical examples and exercises.
L12	SQL Operators Recap	Review key SQL commands, syntax, and concepts such as SELECT, FROM, WHERE, ORDER BY, LIKE, and data structure. Reinforce skills to write and understand basic SQL queries.
L13	JOIN Introduction	Gain an understanding of the definition of JOINS and database normalization and learn why they are needed.
L14	Introduction to Inner JOINs in SQL	Learn how to use INNER JOINs in SQL to combine data from multiple tables, write JOIN queries, and specify columns from different tables using the ON clause.
L15	JOINs in SQL	Learn how to use INNER, LEFT, and RIGHT JOINs in SQL to combine data from multiple tables, including filtering joins and understanding NULLs in join results.
L16	Advanced JOIN Concepts	Explore advanced SQL JOINs, primary and foreign keys, multi-table joins, and aliases to efficiently combine and analyze relational database tables.
L17	JOIN Recap	Recap key concepts learned in this lesson.
L18	SQL Aggregation Introduction	An overview of the content that will be learned in Lesson 3.
L19	NULLs and Aggregation	Learn how NULLs represent missing data in SQL, how they differ from zeros, and how to identify or exclude them using IS NULL or IS NOT NULL in queries and aggregations.

L20	COUNT in SQL	Learn how to use SQL's COUNT function to count rows, understand the difference between COUNT(*) and COUNT(column), and see how NULL values affect results.
L21	SUM in SQL	Learn how to use the SQL SUM function to total numeric columns, handle NULL values, and perform aggregations across data for inventory analysis and sales calculations.
L22	MIN, MAX, and AVG in SQL	Learn to use SQL's MIN, MAX, and AVG aggregate functions to find minimums, maximums, and averages, while understanding their handling of NULLs and practical business applications.
L23	GROUP BY in SQL	Learn how to use GROUP BY in SQL to aggregate data within subsets, group by multiple columns, and combine GROUP BY with ORDER BY for insightful data analysis.
L24	DISTINCT in SQL	Learn how to use DISTINCT in SQL SELECT statements to retrieve unique rows across specified columns, with practical examples and exercises for real-world query scenarios.
L25	HAVING Clause in SQL	Learn how to use the SQL HAVING clause to filter aggregated group results, enabling analysis of grouped data with aggregate functions beyond the WHERE clause.
L26	DATE in SQL	Explore how to use SQL date functions like DATE_TRUNC and DATE_PART to analyze, aggregate, and group data by various date parts for effective reporting.
L27	CASE Statements in SQL	Learn to use SQL CASE statements for conditional logic in SELECT queries, including multi-case logic, handling NULLs, and combining CASE with aggregations.
L28	Introduction to Subqueries	Learn the basics of SQL subqueries, focusing on scalar subqueries, to simplify complex logic and compare values efficiently within a single query.
L29	Subqueries in the WHERE Clause	Learn how to use subqueries in the WHERE clause to filter SQL results dynamically, making queries more flexible and readable. Practice building both scalar and filtering subqueries.
L30	Subqueries in the FROM Clause	Learn how to use subqueries in the FROM clause to create derived tables, transform data, and build complex SQL queries efficiently with practical examples.
L31	Common Table Expressions (CTEs)	Learn how Common Table Expressions (CTEs) make complex SQL queries cleaner, more readable, and efficient by organizing logic into reusable, easily-managed parts with the WITH clause.
L32	Temporary Tables in SQL	Discover how temporary tables in SQL simplify complex queries by storing intermediate results, boosting performance, organizing logic, and supporting efficient data analysis.

L33	Tips for Clean and Efficient SQL	Learn key tips for writing clean, efficient SQL using subqueries, CTEs, and temp tables, plus best practices for clarity, performance, and maintainability.
L34	SQL Window Functions	Discover SQL window functions to perform calculations like totals, averages, and ranking across rows while retaining individual data for advanced analysis.
L35	Working with SQL PARTITION BY	Discover how SQL's PARTITION BY enables advanced analytics—like running totals, rankings, and trend analysis—across groups without losing row-level detail.
L36	Order in Window Functions	Understand the role of ORDER BY in SQL window functions for ordering, ranking, and framing data, including ROWS vs RANGE and practical uses like running totals and moving averages.
L37	Building a Window Function	Learn how to build SQL window functions for detailed analysis, cumulative totals, and efficient queries using PARTITION BY, ORDER BY, and smart optimization strategies.
L38	SQL Data Cleaning Introduction	Learn why data cleaning is crucial, how to identify issues like duplicates, nulls, and outliers, and practical SQL techniques to ensure analysis is accurate and reliable.
L39	Advanced SQL Data Cleaning	Learn advanced SQL data cleaning using staging tables and workflows for scalable, reliable data prep, with best practices for consistency, deduplication, and quality control.
L40	FULL OUTER JOIN	Learn how FULL OUTER JOIN in SQL returns all matched and unmatched rows from both tables, useful for comprehensive data comparison and analysis, with syntax, use cases, and practice.
L41	JOINs with Comparison Operators	Explore SQL JOINs using comparison operators like < and > to connect tables on inequalities, including practical queries and key differences from standard equality joins.
L42	Self JOINs	Learn how to use self JOINs in SQL to compare rows within the same table, often with inequalities, for analyzing event sequences within specific time intervals.
L43	UNION	Learn to use the SQL UNION operator to combine result sets from multiple tables, remove duplicates, and perform further analysis on appended data, including using UNION ALL for all rows.
L44	Performance Tuning	Learn to identify slow SQL queries and improve performance by filtering data, simplifying joins, aggregating before joins, and using EXPLAIN to analyze and optimize query plans.
L45	Congratulations	

L46	Project: Deforestation Exploration	In this project, students will be putting their SQL skills to the test to help determine where to concentrate efforts to combat deforestation.

03. Management of Relational and Non-relational Databases

23 hours

Databases need to be structured properly to enable efficient and effective querying and analysis of data. Build normalized, consistent, and performant relational data models. Use SQL Database Definition Language (DDL) to create the data schemas designed in Postgres and apply SQL Database Manipulation Language (DML) to migrate data from a denormalized schema to a normalized one. Understand the tradeoffs between relational databases and their non-relational counterparts, and justify which one is best for different scenarios. With a radical shift of paradigms, learn about MongoDB and Redis to get an understanding of the differences in behaviors and requirements for non-relational databases.

L1	Intro to DBMS: Relational and Non-Relational Databases	Get introduced to database management systems, as well as learning the difference between relational and non-relational databases.
L2	Normalizing Data	Find out about the different forms of normalized data for optimizing database storage.
L3	Data Definition Language (DDL)	Learn about the data definition language, such as creating tables and different data types.
L4	Data Manipulation Language (DML)	Dive into the data manipulation language in order to alter existing tables and data.
L5	Consistency with Constraints	Find out how to keep consistency amongst your data by adding constraints.
L6	Performance with Indexes	Maximize your database performance by using indexes when and where appropriate.
L7	Intro to Non-Relational Databases	With your relational database skills in hand, you're ready for a look into the other side of data, with non-relational databases.
L8	Project: Udiddit, A Social News Aggregator	Investigate a poorly designed database for Udiddit, a social news aggregator. You'll design a new, normalized and performant database and migrate over data from the previous database.

Meet Your Instructors

Ziad Saab

Software Developer and Co-Founder DecodeMTL

Ziad is a seasoned software developer who loves mentoring and teaching. Currently working as an independent contractor, he previously co-founded and taught full-stack web development at DecodeMTL, Montreal's first web development bootcamp.

David Elliott

Data Scientist, Data Engineer

David Elliott is both a data scientist and a data engineer at a small data management company. He has extensive experience in education, both as an instructor and as a curriculum developer.

Derek Steer

CEO, Superframe

Derek is the CEO at Superframe, and former CEO at Mode. He developed an analytical foundation at Facebook and Yammer and is passionate about sharing it with future analysts.

Malavica Sridhar

Senior Product Manager at CircleUp

Malavica is a Senior Product Manager with over five years of experience. Her work includes building Helio, an ML platform used to identify breakout brands in early-stage consumer packaged goods companies.

Why Udacity

Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- Gain proven experience
- Retain knowledge longer
- Apply new skills immediately

24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- Get help from subject matter experts
- Gain valuable insights and improve your skills
- Learn industry best practices